

MATH3026/4022 Examples Class 2

$$1 \text{ (ii)} \quad X_t = \frac{5}{4} X_{t-1} + \frac{1}{8} X_{t-2} - \frac{3}{8} X_{t-3} + Z_t$$

$$\Rightarrow X_t - \frac{5}{4} X_{t-1} - \frac{1}{8} X_{t-2} + \frac{3}{8} X_{t-3} = Z_t$$

$$\Rightarrow \left(1 - \frac{5}{4}B - \frac{B^2}{8} + \frac{3}{8}B^3\right) X_t = Z_t$$

The process is (weakly) stationary if roots of

$\phi(z) = 0$ lie outside the unit circle.

$$\text{Where } \phi(z) = 1 - \frac{5}{4}z - \frac{z^2}{8} + \frac{3}{8}z^3$$

$$\therefore \text{Set } 1 - \frac{5}{4}z - \frac{z^2}{8} + \frac{3}{8}z^3 = 0.$$

$$\text{Here } 1 - \frac{5}{4} - \frac{1}{8} + \frac{3}{8} = 0 \Rightarrow z=1 \text{ is a root.}$$

$z=1$ does not lie outside the unit circle.

$\therefore$ The process is not (weakly) stationary.

$$(iii) \quad X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

$$\psi_j = \frac{1}{1+j}$$

The process is weakly stationary if $\sum_{j=0}^{\infty} \psi_j^2$ is finite.

Here $\sum_{j=0}^{\infty} \psi_j^2 = \sum_{j=0}^{\infty} \left(\frac{1}{1+j}\right)^2$ } is a finite sum.

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

(b) Mean: $E(X_t) = \sum_{j=0}^{\infty} \psi_j E(Z_{t-j})$

$$= 0.$$

(c) $Var(X_t) = Var\left(\sum_{j=0}^{\infty} \frac{1}{j+1} Z_{t-j}\right)$

$$= \sum_{j=0}^{\infty} \frac{1}{(j+1)^2} Var(Z_{t-j})$$

(independence)

$$= \sigma_z^2 \frac{\pi^2}{6}.$$

Consider $h > 0$

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

$$= \mathbb{E}(X_t X_{t+h})$$

$$= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \psi_j Z_{t-j} \right) \left(\sum_{k=0}^{\infty} \psi_k Z_{t+h-k} \right) \right]$$

$$= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \psi_j \psi_k \mathbb{E}(Z_{t-j} Z_{t+h-k})$$

$$\mathbb{E}(Z_{t-j} Z_{t+h-k}) = 0 \text{ except where}$$

$$-j = h - k$$

$$\Rightarrow k = j + h$$

$$\therefore \gamma(h) = \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \mathbb{E}(Z_{t-j}^2)$$

$$= \sigma_z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h}$$

$$= \sigma_z^2 \sum_{j=0}^{\infty} \frac{1}{(j+1)} \times \frac{1}{(j+h+1)}$$

$$= \sigma_z^2 \sum_{j=0}^{\infty} \left\{ \frac{A}{j+1} + \frac{C}{j+h+1} \right\}$$

Partial fractions : $A = \frac{1}{h}$; $C = -\frac{1}{h}$

$$\therefore \gamma(h) = \frac{\sigma_z^2}{h} \left\{ \sum_{j=0}^{\infty} \frac{1}{j+1} - \sum_{j=0}^{\infty} \frac{1}{j+h+1} \right\}$$

$$= \frac{\sigma_z^2}{h} \left\{ \sum_{j=0}^{\infty} \frac{1}{j+1} - \sum_{k=h}^{\infty} \frac{1}{k+1} \right\}$$

$$= \frac{\sigma_z^2}{h} \sum_{j=0}^{h-1} \frac{1}{j+1}$$

$\therefore$ of weak stationarity

$$\gamma(h) = \frac{\sigma_z^2}{|h|} \sum_{j=0}^{|h|-1} \frac{1}{j+1}.$$

$$\gamma(h) = \begin{cases} \sigma_z^2 \frac{\pi^2}{6} & h=0; \\ \frac{\sigma_z^2}{|h|} \sum_{j=0}^{|h|-1} \frac{1}{j+1} & |h| \geq 1. \end{cases}$$

$$2(a) \quad X_t - (\gamma_1 + \gamma_2) X_{t-1} + \gamma_1 \gamma_2 X_{t-2} = Z_t$$

$$\Rightarrow (1 - (\gamma_1 + \gamma_2)B + \gamma_1 \gamma_2 B^2) X_t = Z_t$$

Stationary if roots of

$$1 - (\gamma_1 + \gamma_2)z + \gamma_1 \gamma_2 z^2 = 0 \quad \text{lie outside the unit circle}$$

$$\Rightarrow (1 - \gamma_1 z)(1 - \gamma_2 z) = 0$$

$$\Rightarrow z = \frac{1}{\gamma_1} \text{ or } z = \frac{1}{\gamma_2}$$

Since $|\gamma_1| < 1$ and $|\gamma_2| < 1$

it follows that $\left| \frac{1}{\gamma_1} \right| > 1$; $\left| \frac{1}{\gamma_2} \right| > 1$

$$(b) \quad (1 - \gamma_1 B)(1 - \gamma_2 B) X_t = Z_t$$

$$\Rightarrow X_t = \frac{Z_t}{(1 - \gamma_1 B)(1 - \gamma_2 B)}$$

$$= \left[\frac{A}{1 - \gamma_1 B} + \frac{C}{1 - \gamma_2 B} \right] Z_t$$

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partial fractions.

$$A = \frac{-\gamma_1}{\gamma_2 - \gamma_1}; \quad C = \frac{\gamma_2}{\gamma_2 - \gamma_1}$$

$$\therefore X_t = \frac{1}{\gamma_2 - \gamma_1} \left\{ \frac{-\gamma_1}{1 - \gamma_1 B} + \frac{\gamma_2}{1 - \gamma_2 B} \right\} Z_t$$

$$(1 - \gamma_1 B)^{-1} = \sum_{j=0}^{\infty} \gamma_1^j B^j$$

$$(1 - \gamma_2 B)^{-1} = \sum_{j=0}^{\infty} \gamma_2^j B^j$$

$$\begin{aligned} \therefore X_t &= \frac{1}{\gamma_2 - \gamma_1} \left\{ \sum_{j=0}^{\infty} \gamma_2^{j+1} B^{j+1} - \sum_{j=0}^{\infty} \gamma_1^{j+1} B^j \right\} Z_t \\ &= \frac{1}{\gamma_2 - \gamma_1} \sum_{j=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) Z_{t-j} \end{aligned}$$

$$(c) \mathbb{E}(X_t X_{t+h})$$

$$= \mathbb{E} \left[\frac{1}{\gamma_2 - \gamma_1} \sum_{j=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) Z_{t-j} \right.$$

$$\times \left. \frac{1}{\gamma_2 - \gamma_1} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1}) Z_{t+h-k} \right]$$

$$= \frac{1}{(\gamma_2 - \gamma_1)^2} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) (\gamma_2^{k+1} - \gamma_1^{k+1}) \times \mathbb{E}(Z_{t-j} Z_{t+h-k}).$$

$$\mathbb{E}(Z_{t-j} Z_{t+h-k}) = 0 \text{ except where } k = j+h$$

$$\therefore \gamma(h) = \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \sum_{j=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) (\gamma_2^{0+h+1} - \gamma_1^{j+h+1})$$

$$= \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \left\{ \begin{aligned} &\sum_{j=0}^{\infty} \gamma_2^{2j+2+h} \\ &- \sum_{j=0}^{\infty} \gamma_1^{j+h+1} \gamma_2^{j+1} \\ &- \sum_{j=0}^{\infty} \gamma_1^{j+1} \gamma_2^{j+h+1} \\ &+ \sum_{j=0}^{\infty} \gamma_1^{2j+h+1} \end{aligned} \right\} \quad \left. \begin{array}{l} \text{sums to} \\ \infty \text{ of} \\ \text{geometric} \\ \text{series} \end{array} \right\} \quad \left[\sum_{j=0}^{\infty} ar^j = \frac{a}{1-r} \right]$$

$$\gamma(h) = \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \left[\begin{aligned} &\frac{\gamma_2^{h+2}}{1 - \gamma_2^2} - \frac{\gamma_1 \gamma_2^{h+1}}{1 - \gamma_1 \gamma_2} \\ &- \frac{\gamma_2 \gamma_1^{h+1}}{1 - \gamma_1 \gamma_2} + \frac{\gamma_1^{h+2}}{1 - \gamma_1^2} \end{aligned} \right]$$

$$5. \text{ If } X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \bar{z}_t$$

then this is (weakly) stationary if

$$\phi_1 + \phi_2 < 1 ; \quad \phi_2 - \phi_1 < 1 ; \quad |\phi_2| < 1 \quad (L4)$$

Let's assume these hold.

For an AR(2) process

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} \quad \text{and} \quad \rho(2) = \phi_2 + \frac{\phi_1^2}{1 - \phi_2} \quad (L4)$$

$$\text{Since } |\phi_2| < 1 \Rightarrow 1 - \phi_2 > 0$$

$$|\rho(1)| = \frac{|\phi_1|}{1 - \phi_2} \quad (*)$$

$$\text{Since } \phi_1 + \phi_2 < 1 \Rightarrow \phi_1 < 1 - \phi_2$$

$$\therefore \text{In } (*) \quad |\rho(1)| = \frac{|\phi_1|}{1 - \phi_2} < \frac{|1 - \phi_2|}{1 - \phi_2} = 1$$

$$\therefore |\rho(1)| < 1$$

$$\rho(2) = \phi_2 + \frac{\phi_1^2}{1 - \phi_2}$$

$$= \phi_2 + (1 - \phi_2) \rho(1)^2 \quad (**)$$

$$e(2) = e(1)^2 + \phi_2 (1 - e(1)^2)$$

$$> e(1)^2 - (1 - e(1)^2) \quad \text{because} \\ \phi_2 > -1$$

$$\Rightarrow e(2) > 2e(1)^2 - 1$$

$$\Rightarrow e(1)^2 < \frac{1}{2} [e(2) + 1]$$

$$\text{Because } e(1)^2 \geq 0$$

$$\Rightarrow 0 < \frac{1}{2} [e(2) + \frac{1}{2}]$$

$$e(2) > -1$$

$$\text{From } (*) \quad e(2) < 1$$

$$\therefore -1 < e(2) < 1.$$

$$6. \quad X_t = \frac{2}{3} X_{t-1} - \frac{2}{9} X_{t-2} + Z_t$$

$$\Rightarrow X_t - \frac{2}{3} X_{t-1} + \frac{2}{9} X_{t-2} = Z_t$$

(a) Y-W eq'ns

$$e(h) - \frac{2}{3} e^{(h-1)} + \frac{2}{9} e^{(h-2)} = 0$$

(b) Auxiliary equation

$$\lambda^2 - \frac{2}{3} \lambda + \frac{2}{9} = 0$$

$$\left(\lambda - \frac{1}{3}\right)^2 + \frac{1}{9} = 0$$

$$\Rightarrow \left(\lambda - \frac{1}{3}\right)^2 = -\frac{1}{9}$$

$$\Rightarrow \lambda - \frac{1}{3} = \pm \frac{i}{3}$$

$$\Rightarrow \lambda = \frac{1}{3} (1 \pm i)$$

$$= \frac{\sqrt{2}}{3} e^{\pm i\frac{\pi}{4}}$$

$\therefore$ The general solution has the form

$$e(h) = C_1 \left(\frac{\sqrt{2}}{3}\right)^{|h|} e^{i\frac{\pi}{4}|h|} + C_2 \left(\frac{\sqrt{2}}{3}\right)^{|h|} e^{-i\frac{\pi}{4}|h|}$$

$$e(0) = 1 \Rightarrow C_1 + C_2 \quad (1)$$

$$e(1) - \frac{2}{3} e(0) + \frac{2}{9} e(-1) = 0$$

$$\Rightarrow \frac{11}{9} e(1) = \frac{2}{3}$$

$$\Rightarrow e(1) = \frac{6}{11}$$

$$C_1 \left(\frac{\sqrt{2}}{3} \right) e^{i\pi/4} + C_2 \left(\frac{\sqrt{2}}{3} \right) e^{-i\pi/4} = \frac{6}{11}$$

$$\Rightarrow C_1 \left(\frac{\sqrt{2}}{3} \right) e^{i\pi/4} + (1 - C_1) \left(\frac{\sqrt{2}}{3} \right) e^{-i\pi/4} = \frac{6}{11}$$

$$C_1 \left(\frac{\sqrt{2}}{3} \right) (e^{i\pi/4} - e^{-i\pi/4}) = \frac{6}{11} - \left(\frac{\sqrt{2}}{3} \right) e^{-i\pi/4}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$C_1 \left(\frac{\sqrt{2}}{3} \right) (2i \sin(\frac{\pi}{4})) = \frac{6}{11} - \left(\frac{\sqrt{2}}{3} \right) e^{-i\pi/4}$$

$$C_1 \left(\frac{\sqrt{2}}{3} \right) 2i \frac{1}{\sqrt{2}} = \frac{6}{11} - \frac{\sqrt{2}}{3} \left(\frac{1}{\sqrt{2}} (1-i) \right)$$

$$2i \frac{C_1}{3} = \frac{6}{11} - \frac{1}{3} (1-i)$$

$$\Rightarrow C_1 = \frac{1}{2} - \frac{7i}{22}$$

$$\therefore c_2 = \frac{1}{2} + \frac{7i}{22}$$

$$e(h) = \left(\frac{1}{2} - \frac{7i}{22} \right) \left(\frac{\sqrt{2}}{3} \right)^{|h|} e^{i\pi|h|/4}$$

$$+ \left(\frac{1}{2} + \frac{7i}{22} \right) \left(\frac{\sqrt{2}}{3} \right)^{|h|} e^{-i\pi|h|/4}$$

$$= \sqrt{\frac{1}{2}}$$

$$= \left(\frac{\sqrt{2}}{3} \right)^{|h|} \left[\frac{1}{2} (e^{i\pi|h|/4} + e^{-i\pi|h|/4}) \right. \\ \left. + \frac{7i}{22} (e^{i\pi|h|/4} - e^{-i\pi|h|/4}) \right]$$

$$= \left(\frac{\sqrt{2}}{3} \right)^{|h|} \left[\cos\left(\frac{\pi|h|}{4}\right) \right. \\ \left. + \frac{7}{11} \sin\left(\frac{\pi|h|}{4}\right) \right].$$

$$h \geq 0.$$

$$\text{or } h < 0.$$

7. (a) MA(1) process

- Sample ACF appears to cut off to zero after lag 1.
- Conversely, the sample PACF does not appear to cut off to zero after a particular lag. Instead, this declines gradually.

(b) Suppose we want to fit an MA(1) model:

$$X_t = \theta Z_{t-1} + Z_t$$

We wish to estimate θ and σ_Z^2 .

For the MA(1) process

$$\rho(1) = \frac{\theta}{1 + \theta^2} \quad \text{[Autocovariance gf]}$$

Looking at the plot, the sample ACF at lag 1 is approximately -0.4 .

$$\therefore \text{Set } \rho(1) = r_1$$

$$\Rightarrow \frac{\tilde{\theta}}{1 + \tilde{\theta}^2} = -0.4$$

$$\Rightarrow \tilde{\theta} = -0.4(1 + \tilde{\theta}^2)$$

$$\Rightarrow 0.4\tilde{\theta}^2 + \tilde{\theta} + 0.4 = 0$$

$$\tilde{\theta}^2 + \frac{5}{2} \tilde{\theta} + 1 = 0$$

$$\left(\tilde{\theta} + \frac{5}{4}\right)^2 - \frac{9}{16} = 0$$

$$\left(\tilde{\theta} + \frac{5}{4}\right)^2 = \frac{9}{16}$$

$$\tilde{\theta} + \frac{5}{4} = \pm \frac{3}{4}$$

$$\tilde{\theta} = -\frac{1}{2} \quad \text{or} \quad \tilde{\theta} = -2$$

We choose $\tilde{\theta} = -\frac{1}{2}$ so that the process is invertible.

~~$\tilde{\sigma}_z^2 = s^2$~~ (s^2 is the sample variance)

$$\text{Var}(X_t) = s^2$$

$$\Rightarrow \tilde{\sigma}_z^2 (1 + \tilde{\theta}^2) = s^2$$

From the time plot, 'most' observations lie within $\pm \frac{5}{2}$ i.e. in the interval $(-\frac{5}{2}, \frac{5}{2})$
s.d. of $\{X_t\}$

set $2s = \frac{5}{2}$

$$\Rightarrow s = \frac{5}{4}$$

$$\therefore \tilde{\sigma}_z^2 \left(1 + \left(-\frac{1}{2}\right)^2\right) = \frac{25}{16} \Rightarrow \tilde{\sigma}_z^2 = \frac{5}{4}$$